

University Interscholastic League
2024 – 2025 Junior High Number Sense Test A

Contestant's Number _____

Final		
2 nd		
1 st		
	Score	Initials

**Read Directions Carefully
Before Beginning Test**

**Do Not Unfold This Sheet
Until Told to Begin**

Directions: Do not turn this page until the person conducting this test gives the signal to begin. This is a ten-minute test. There are 80 problems. Solve accurately and quickly as many as you can in the order in which they appear. **ALL PROBLEMS ARE TO BE SOLVED MENTALLY.** Make no calculations with paper and pencil. Write only the answer in the space provided at the end of each problem. Problems marked with a (*) require approximate integral answers; any answer to a starred problem that is within five percent of the exact answer will be scored correct; all other problems require exact answers.

The person conducting this contest should explain these directions to the contestants.

Stop – Wait for Signal!

- | | |
|---|---|
| <p>(1) $2025 \div 25 =$ _____</p> <p>(2) $25 \times 16 =$ _____</p> <p>(3) $103 - 59 =$ _____</p> <p>(4) $1101 \div 3 =$ _____</p> <p>(5) $\frac{7}{12} - \frac{1}{4} =$ _____</p> <p>(6) $24 \div 0.25 =$ _____</p> <p>(7) $38 \times 11 =$ _____</p> <p>(8) $17^2 =$ _____</p> <p>(9) $24 - 16 \div 4 + 2 =$ _____</p> <p>*(10) $2024 \times 99 + 2025 =$ _____</p> <p>(11) $19 \times 21 =$ _____</p> <p>(12) The median of 18, 20, 15, 12 is _____</p> <p>(13) Which is smaller: $\frac{9}{11}$ or $\frac{11}{13}$? _____</p> <p>(14) $21 \times 18 + 9 \times 18 =$ _____</p> <p>(15) $38 \times 78 =$ _____</p> <p>(16) $220 - 31 - 29 =$ _____</p> <p>(17) $27^2 =$ _____</p> <p>(18) $17 + 20 + 23 + 26 + 29 =$ _____</p> <p>(19) MDXL = _____ (Arabic Numeral)</p> | <p>*(20) $361 \times 330 =$ _____</p> <p>(21) If $2x + 13 = 21$, then $x^2 =$ _____</p> <p>(22) $102 \times 104 =$ _____</p> <p>(23) $9\frac{5}{12} - 6\frac{2}{3} =$ _____</p> <p>(24) A $12\frac{1}{2}\%$ sales tax on \$16 is \$ _____</p> <p>(25) $3\frac{2}{3} \times 3\frac{1}{3} =$ _____ (mixed number)</p> <p>(26) $2.5 \times 4.4 \times 10 =$ _____</p> <p>(27) 1 gallon = _____ cubic inches</p> <p>(28) The GCD of 18, 12 and 20 is _____</p> <p>(29) How many whole numbers evenly divide 24? _____</p> <p>*(30) $545 \times 121 - 45 =$ _____</p> <p>(31) $37 \times 73 =$ _____</p> <p>(32) If a square, with area 36, and equilateral triangle have the same perimeter, the triangle side = _____</p> <p>(33) If $1.75 - 0.5 = n$, the $n^{-1} =$ _____</p> <p>(34) $-5^2 =$ _____</p> <p>(35) 3 cups = _____ ounces</p> <p>(36) 1 inch = _____ centimeters</p> <p>(37) $41\frac{2}{3}\%$ of 36 is _____</p> |
|---|---|

- (38) $111 \times 678 =$ _____
- (39) The area of a square with a diagonal of 8 is _____
- *(40) 410 yards = _____ inches
- (41) For the sequence: 3, **n**, 11, **q**, 19, 23, . . . **n + q** equals _____
- (42) $5^4 + 5^3 =$ _____
- (43) $75 \times 23 =$ _____
- (44) The number of elements in $\{1, 2, 3, 5\} \cup \{2, 4, 6, 8\}$ is _____
- (45) $4 \text{ (base 5)} \times 41 \text{ (base 5)} =$ _____ (base 5)
- (46) The length of the edge of a cube with surface area 150 is _____
- (47) The perimeter of a right triangle with leg 5 and hypotenuse 13 is _____
- (48) The largest prime number less than 100 is _____
- (49) The volume of a rectangular solid that measures 8 by 5 by x is 120. What is x ? _____
- *(50) $39782 \div 25 =$ _____
- (51) What is the area of the trapezoid with bases $7\frac{1}{2}$, $9\frac{1}{2}$ and height 8? _____
- (52) What is the 10th term in the sequence: 1, 3, 5, . . .? _____
- (53) $101 \times 468 =$ _____
- (54) Twice a number added to 12 is 14. The number is _____
- (55) What is the perimeter of a rectangle with diagonal 13 and length 12? _____
- (56) The area of a rhombus with diagonals 12 and x is 60. What is x ? _____
- (57) $48 \times 24 =$ _____
- (58) $(12^2 + 8 \times 6) \div 5$ has a remainder of _____
- (59) If $13 - 2x < 27$, then $x >$ _____
- *(60) $35^3 =$ _____
- (61) $(-12) - (-24) \div (-3) =$ _____
- (62) $25\frac{1}{3}$ days = _____ hours
- (63) 20 meters/sec = _____ km/hr
- (64) $0.4666\ldots =$ _____ (common fraction)
- (65) $\frac{3}{4} + \frac{4}{3} =$ _____
- (66) The volume of a circular cylinder with radius 10 and length 25 is $k\pi$. What is k ? = _____
- (67) The sum of the whole number divisors of 12 is _____
- (68) $3! \times 4! =$ _____
- (69) 225% of 444 = _____
- *(70) 15 square miles = _____ acres
- (71) $286 \times 49 =$ _____
- (72) $43\frac{3}{4}\% =$ _____ (common fraction)
- (73) $22 \times 0.8181\ldots =$ _____
- (74) The surface area of a sphere with a diameter of 8 is $k\pi$ and $k =$ _____
- (75) What is the probability of drawing a red queen from a standard deck of 52 cards? _____
- (76) If the shortest distance between the points (6, 5) and (0, **x**) is 10, what is **x**? _____
- (77) $5^3 \div 25 =$ _____
- (78) 18% of **n** equals 9% of 20. **n** = _____
- (79) $20 + 19 + 18 + \ldots + 0 =$ _____
- *(80) $749 \times 359 =$ _____

2024 – 2025 University Interscholastic League Junior High Number Sense Test A – Key

(1) 81	*(20) 113174 – 125086	(38) 75258	(59) -7
(2) 400	(21) 16	(39) 32	*(60) 40732 – 45018
(3) 44	(22) 10608	*(40) 14022 – 15498	(61) -20
(4) 367	(23) $2\frac{3}{4}$; 2.75; $\frac{11}{4}$	(41) 22	(62) 608
(5) $\frac{1}{3}$	(24) 2.00	(42) 750	(63) 72
(6) 96	(25) $12\frac{2}{9}$	(43) 1725	(64) $\frac{7}{15}$
(7) 418	(26) 110	(44) 7	(65) $2\frac{1}{12}$; $\frac{25}{12}$
(8) 289	(27) 231	(45) 314	(66) 2500
(9) 22	(28) 2	(46) 5	(67) 28
*(10) 192281 – 212521	(29) 8	(47) 30	(68) 144
(11) 399	*(30) 62605 – 69195	(48) 97	(69) 999
(12) 16.5 ; $16\frac{1}{2}$; $\frac{33}{2}$	(31) 2701	*(50) 1512 – 1670	*(70) 9120 – 10080
(13) $\frac{9}{11}$	(32) 8	(51) 68	(71) 14014
(14) 540	(33) $\frac{4}{5}$; .8	(52) 19	(72) $\frac{7}{16}$
(15) 2964	(34) -25	(53) 47268	(73) 18
(16) 160	(35) 24	(54) 1	(74) 64
(17) 729	(36) 2.54 ; $2\frac{27}{50}$; $\frac{127}{50}$	(55) 34	(75) $\frac{1}{26}$
(18) 115	(37) 15	(56) 10	(76) -3
(19) 1540		(57) 1152	(77) 5
		(58) 2	(78) 10
			(79) 210
			*(80) 255447 – 282335

Note: *(Number) x – y means an integer between x and y inclusive.

If an answer is of the type like $\frac{2}{3}$ it cannot be written as .666... or $\overline{.6}$.